RAG System Testing Knowledge Document

Section 1: Artificial Intelligence Research

Artificial Intelligence research began formally in **1956 at the Dartmouth Conference**, which was organized by **John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon**. The conference introduced the term **Artificial Intelligence**.

A famous AI program called **ELIZA** was created in **1966 by Joseph Weizenbaum at MIT**. ELIZA simulated a psychotherapist and demonstrated how machines could mimic conversation.

Another important AI milestone was **IBM's Deep Blue**, which defeated world chess champion **Garry Kasparov in 1997**.

In **2016**, the AI system **AlphaGo**, developed by **DeepMind**, defeated the world champion **Lee Sedol** in the board game Go.

Section 2: Machine Learning Algorithms

Machine Learning algorithms can be categorized into **three main types**:

1. **Supervised Learning**
2. **Unsupervised Learning**
3. **Reinforcement Learning**

The **Support Vector Machine (SVM)** algorithm was introduced by **Vladimir Vapnik and Alexey Chervonenkis**.

The **Naive Bayes algorithm** is based on **Bayes' Theorem**, which was formulated by **Thomas Bayes in the 18th century**.

The **Random Forest algorithm** was proposed by **Leo Breiman in 2001**.

Section 3: Deep Learning Models

Deep Learning models are built using **artificial neural networks with multiple hidden layers**.

The **AlexNet model**, created by **Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton**, won the **ImageNet competition in 2012**.

The **Transformer architecture** was introduced in **2017** in the paper titled **"Attention Is All You Need"** by researchers from **Google Brain**, including **Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, and Illia Polosukhin**.

Transformers rely heavily on a mechanism called **self-attention**, which helps models understand relationships between words in a sentence.

Section 4: Large Language Models

Large Language Models (LLMs) are trained on large text datasets and use the transformer architecture.

Examples include:

- **GPT-3**, released by **OpenAI in 2020**, with **175 billion parameters**
- **LLaMA**, released by **Meta AI in 2023**
- **PaLM**, released by **Google in 2022**, with **540 billion parameters**

These models are used in applications such as **chatbots, code generation, translation, and document question answering systems**.

Section 5: Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a technique that combines **information retrieval** with **language generation**.

A typical RAG pipeline contains the following components:

1. **Document Loader**
2. **Text Splitter**
3. **Embedding Model**
4. **Vector Database**
5. **Retriever**
6. **Large Language Model**

Vector databases such as **FAISS, Pinecone, Chroma, and Weaviate** are commonly used.

Embeddings convert text into **high-dimensional numerical vectors**. These vectors allow the system to measure **semantic similarity** between different pieces of text.

Section 6: Computer Vision Dataset Example

Consider a medical image dataset used for skin disease detection.

The dataset contains the following categories:

- **Monkeypox — 279 images**

- **Chickenpox — 107 images**
- **Measles — 91 images**
- **Normal skin — 293 images**

The dataset therefore contains a total of **770 images**.

A convolutional neural network trained on this dataset achieved **90% classification accuracy**.